

Medical Robotics

Prof. Marilena Vendittelli
Sapienza Università di Roma

Lecture 12

Teleoperation

Principles, Stability and Transparency
in Bilateral Teleoperation

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1. Lecture Overview

This lecture introduces the fundamental principles of **teleoperation** in the context of medical robotics. Starting from terminology (teleoperation, telepresence, telerobotics), the lecture covers the main components of a teleoperated system, a brief historical overview, the peculiarities of telemanipulators, and the three main control modalities. The central portion is devoted to **bilateral direct control**: kinematic coupling strategies, the role of scaling factors in shaping kinesthetic perception, and the fundamental trade-off between **transparency** and **stability**. The lecture concludes by deriving the dynamic models of the human operator, the remote environment, and both the master and slave manipulators — expressed in Cartesian space after feedback linearisation — leading to the unified single-axis linear model used in subsequent analyses.

Connection to Lecture 13. The system-level model derived here will be extended using the **two-port network** formalism to formally characterise stability and transparency, and to compare concrete control architectures for bilateral teleoperation.

2. Motivation: Teleoperation in Medical Robotics

Teleoperation is of particular interest in medical robotics because teleoperated systems appear in nearly every clinical domain. With the notable exception of orthopaedics — where autonomous robotic milling is standard — most surgical robots operate in teleoperation mode. The Da Vinci surgical system is the prime example, extending the surgeon's dexterity and perception into the patient's body through a master–slave architecture.

3. Terminology

Teleoperation. Extension of a person's sensing and manipulating capability to a *remote location*. The system includes not only the manipulator but also sensors, actuators, and a communication channel to and from the operator.

Telepresence. The operator *feels* physically present at the remote site; the dexterity of the remote device matches that of the bare-hand human operator. We will characterise this concept mathematically in Section 9.

Telerobotics. Remote control of robots; a combination of teleoperation and telepresence. The term implies the use of robotic devices (not merely passive pantographs) on both local and remote sides.

4. Main Components of a Teleoperated System

A **telem manipulator** is a complex electro-mechanical system encompassing:

- A **master (local) device** together with its controller.
- A **slave (remote) device** together with its controller.
- A **communication channel** between master and slave.

The system is interfaced on the master side with a *human operator* and on the slave side with the *environment*. This two-sided interaction is the defining feature of bilateral teleoperation.

The complexity of the two devices can differ enormously: for example, the master could be as simple as a joystick while the slave is a full humanoid robot — a common situation in

space or rescue applications. When devices are very different, the terminology “master/slave” is increasingly replaced by “local/remote”; both are used interchangeably in this course.

5. A Brief History

Teleoperation predates medical robotics by nearly two centuries:

- **Early 1800s.** Thomas Jefferson used (and may have adapted) the *polygraph*, a four-bar linkage (pantograph) that reproduced his signature in real time on a second sheet of paper — a fully mechanical, zero-latency, one-to-one teleoperation system.
- **1940s.** Raymond Goertz at Argonne National Laboratory developed mechanical pantographs for remote handling of radioactive materials. Master and slave were mechanically identical; coupling was purely mechanical at the joint level. No actuation or sensing electronics were involved.
- **1954.** Electromechanical manipulators with feedback servo control were introduced, enabling master and slave to be *physically different* devices for the first time.
- **1960s.** Closed-circuit television (CCTV) and head-mounted displays allowed operators to be immersed in a remote visual scene — the beginning of modern teleoperation.

Contemporary applications span space robotics (Luca Parmitano tele-operating an arm-equipped rover from the International Space Station during the Analog-1 campaign), subsea robotics (Ocean One at Stanford), military systems, and — most relevant to this course — **surgical robotics** (Da Vinci).

6. Peculiarities of Telemanipulators

Relative to a classical autonomous industrial robot, a telemanipulator has several distinctive characteristics:

1. **Human in the control loop.** The operator provides high-level intelligence and adaptability. Information fed back to the human must be *immediately and naturally interpretable* to avoid increasing cognitive load.
2. **Slave-to-master feedback.** The human must receive information on exchanged forces, slave position, video, and potentially tactile or acoustic signals.
3. **Presence of a communication channel.** Commands travel over a channel that may introduce **delays** and has **limited bandwidth**. The choice of which signals to transmit and at what rate is non-trivial.
4. **Kinematic heterogeneity.** Since the 1950s it has been common for master and slave to differ in kinematics, dimensions, and workspace. Methods to map commands from one to the other (*motion retargeting*) are required.
5. **Unstructured environment.** The remote robot interacts with an environment whose physical properties (friction, mass, impedance, disturbances) are largely unknown — especially in surgical applications where tissue properties vary between patients and within a single procedure.

7. Control Modalities

Depending on the type of connection between master and slave, control architectures fall into three main categories:

Direct (bilateral) control. Commands are transmitted from master to slave; force and motion information is fed back from slave to master. No local intelligence is assumed at the slave side. *This is the focus of this lecture.*

Shared control. Local sensory feedback loops at the teleoperator site allow the slave robot to refine gross operator commands autonomously, giving the teleoperator a modest degree of intelligence. Useful when (a) large communication delays require the slave to act independently between command refreshes, or (b) motion compensation for physiological movement is needed.

Supervisory control. The remote robot operates with full autonomy; the human supervises and can override when necessary.

Shared control example: beating-heart surgery. A local controller synchronises the slave end-effector with cardiac motion (using visual servoing, proposed in the 1990s and tested on animals), so the surgeon perceives a stationary tissue. The limited mechanical bandwidth of available robots (too much delay to track the heartbeat faithfully) prevented clinical adoption. For slower physiological motions such as respiration the concept is more tractable.

8. Direct Control and Bilateral Teleoperation

8.1. Kinematic Coupling: Overview

Two manipulators must be *connected* so that the slave follows the master. Issues to resolve at the outset:

- **Initial offset:** the two devices generally start in different configurations; this offset must be compensated in the reference signals.
- **Workspace mismatch:** if the master's workspace is smaller, it must be periodically disconnected from the slave and repositioned (*clutching*).

8.2. Joint-Level Coupling (Kinematically Equivalent Devices)

When master and slave are kinematically identical (or equivalent up to a constant scale), direct joint coupling is possible. Let subscripts m , s and r denote master, slave, and reference (desired value) respectively, and define:

$$\mathbf{q}_{\text{offset}} = \mathbf{q}_m^{(0)} - \mathbf{q}_s^{(0)}$$

where the superscript (0) denotes the initial configuration. The reference signals are then:

$$\mathbf{q}_s^r = \mathbf{q}_m + \mathbf{q}_{\text{offset}}, \quad \mathbf{q}_m^r = \mathbf{q}_s - \mathbf{q}_{\text{offset}}.$$

The slave tracks the master's joint values (offset-corrected). The master is simultaneously given the slave's current position as its own reference so that it can slow down if the slave lags.

8.3. Task-Level Coupling (Kinematically Dissimilar Devices)

When master and slave differ in kinematics — the most common situation in surgical teleoperation — coupling is performed in Cartesian (task) space. Using $\mathbf{x}$ for Cartesian position:

$$\mathbf{x}_s^r = \mathbf{x}_m + \mathbf{x}_{\text{offset}}, \quad \mathbf{x}_m^r = \mathbf{x}_s - \mathbf{x}_{\text{offset}}.$$

For **orientation**, expressed via rotation matrices, an additive offset is replaced by a multiplicative one:

$$R_s^r = R_m R_{\text{offset}}, \quad R_m^r = R_s R_{\text{offset}}^{-1}, \quad R_{\text{offset}} = R_m^{-1}|_0 \cdot R_s|_0.$$

If master and slave start with identical orientations, $R_{\text{offset}} = I$ (no correction needed). This formulation avoids singularities and ambiguities that arise with Euler-angle or quaternion differences.

8.4. Clutching

When the master workspace is smaller than the slave workspace, the operator must:

1. **Disconnect** (disengage) the master from the slave.
2. **Reposition** the master within its own workspace (e.g., return a joystick to centre).
3. **Reconnect** and continue issuing commands from the new master position.

This is directly analogous to lifting a computer mouse and re-placing it. Clutching is implemented in all surgical teleoperation systems including the Da Vinci.

8.5. Visual and Ergonomic Frame Convention

When a camera at the remote site feeds a monitor at the local site, a natural coupling requires:

- Slave position and orientation measured *relative to the camera frame*.
- Master position and orientation measured *relative to the operator's viewpoint*.

This is the convention adopted by the Da Vinci system: the surgeon has the impression that the remote instruments are directly under his or her gaze.

9. Scaling Factors and Kinesthetic Perception

9.1. Why Scale?

Different manipulators and interacting environments require scaling of physical characteristics. Crucially, different types of information about the same environment scale with *different* factors. If the environment is magnified by a factor K :

Quantity	Scale factor	Example
Linear dimension (position, length)	K	Distance to target
Area / surface	K^2	Contact patch
Volume / mass	K^3	Tissue volume

9.2. Scaling Factors for Kinesthetic Perception

The kinesthetic perception — the perception of force *versus* motion, i.e. the impedance of the remote environment — is shaped by two scalar factors:

$$\dot{x}_s = \lambda_v \dot{x}_m, \quad f_m = \lambda_f f_s$$

λ_v : velocity scaling factor (slave velocity relative to master velocity).

λ_f : force scaling factor (force fed back to master relative to slave–environment force).

The impedance relationship $f = Z \dot{x}$ and the instantaneous power $P = f \dot{x}$ yield the following relations between master and slave:

$$\boxed{P_m = \lambda_f / \lambda_v P_s}, \quad \boxed{Z_m = \lambda_v \lambda_f Z_s}$$

Among the four port variables $(\dot{x}_m, f_m, \dot{x}_s, f_s)$, **only three independent relationships** can be freely assigned; the fourth is determined by the other three. Two key observations follow:

- **Fixing the ratio** λ_f / λ_v prescribes the power relationship P_m / P_s .
- **Fixing the product** $\lambda_f \cdot \lambda_v$ prescribes the impedance relationship Z_m / Z_s .
- **Both cannot be fixed simultaneously** (and independently chosen) unless $\lambda_v = \lambda_f = 1$.

Telepresence — mathematical definition. The condition $\lambda_v = \lambda_f = 1$ realises a *dynamic similarity* between master and slave: the operator perceives exactly the impedance of the remote environment ($Z_m = Z_s$) and the power exchanged is identical on both sides ($P_m = P_s$). This is the ideal of complete transparency. As it implies both power and impedance matching simultaneously, it is the only point where full transparency and passivity coexist — but only in the zero-delay case.

9.3. Worked Example: Surgical Scaling with $\lambda_v = \lambda_f = 0.2$

Consider a Da Vinci-like system with $\lambda_v = \lambda_f = 0.2$.

Effect on power.

$$P_m = P_s$$

Same energy exchanged at both ports.

Effect on impedance.

$$Z_m = \lambda_v \lambda_f Z_s = 0.04 Z_s$$

The surgeon perceives a softer tissue than it actually is.

Clinical consequences.

Effect	Sign	Reason
Tissue perceived as softer	✓	Reduces risk of accidental damage
Slave motion 5× smaller than master	✓	Higher spatial precision
Perception not transparent	×	$Z_m \neq Z_s$; palpation unreliable

9.4. Transparent Scaling and Why It is Dangerous

For a realistic impedance perception one would like $Z_m = Z_s$, which requires $\lambda_v \lambda_f = 1$, e.g. $\lambda_v = 1/\lambda_f$. Then:

$$P_m = \lambda_f^2 P_s$$

Power amplification can cause instability in the presence of a communication delay.

Fundamental trade-off. Transparency (faithful impedance reproduction) and stability (boundedness / passivity) are *conflicting* requirements. The common surgical choice $\lambda_v = \lambda_f < 1$ sacrifices some transparency in favour of power preservation and hence stability. Choosing $\lambda_v = \lambda_f$ also means the ratio $\lambda_f / \lambda_v = 1$, i.e. the *power* relationship is controlled.

Controlling power (energy) is the most robust guarantee of stability; controlling impedance alone is not.

10. Teleoperation System Architecture

At the system level a bilateral teleoperation system comprises four dynamic subsystems, as summarised below.

Subsystem	Description
Human operator	Dynamic system on the local side; source of intentional commands. Interacts with the master device exchanging force f_H and position x_m .
Master manipulator	Local robot; moves under the action of the human force f_H and the control force f_{Mc} .
Slave manipulator	Remote robot; moves under the action of the control force f_{Sc} and the environment reaction force f_E .
Remote environment	Passive dynamic system (tissue, object) at the slave side; characterised by visco-elastic dynamics.

The communication channel connects the master controller to the slave controller. **Zero communication delay** is assumed in the analysis of this and the next lecture; handling delayed systems is an active open research area.

11. Dynamic Models of the Subsystems

11.1. Human Operator

The force f_H exchanged between the human hand and the master handle is the net difference between the *intentional force* f_H^* and the force due to the *impedance of the human arm*. Assuming a linear model (valid locally):

$$f_H = f_H^* - Z_H(s) \dot{x}_H$$

where $Z_H(s) = M_H s + B_H + K_H/s$ is the Laplace-domain arm impedance (mass, damping, stiffness). Since hand and handle move together: $x_H = x_m$.

Measuring or estimating f_H^* independently is very difficult; the split into intentional force and arm impedance is a modelling choice supported by experimental evidence but not directly observable.

11.2. Remote Environment

The environment is modelled as a linear second-order system:

$$M_e \ddot{x}_e + B_e \dot{x}_e + K_e (x_e - \bar{x}_e) = f_e^* - f_E$$

where:

- M_e, B_e, K_e : inertia, damping, stiffness of the environment;
- $\bar{x}_e$: rest (equilibrium) position before contact;
- f_e^* : exogenous active force (e.g., heartbeat or respiration — treated as zero in the nominal model, as a disturbance otherwise);
- f_E : interaction force exerted by the slave on the environment.

During contact: $x_e = x_s$. Before contact: $f_E = 0$.

The second-order linear model is a reasonable approximation for *low-frequency* tissue interactions. At higher frequencies, nonlinear and viscoelastic effects become significant and more complex models (e.g., Kelvin–Voigt or Hunt–Crossley contact models) are required.

11.3. Manipulator Dynamics in Joint Space

The standard rigid-body joint-space model for free motion is:

$$M(\mathbf{q}) \ddot{\mathbf{q}} + C(\mathbf{q}, \dot{\mathbf{q}}) \dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) = \mathbf{u}$$

where M is the inertia matrix, C encodes Coriolis and centrifugal terms, $\mathbf{g}$ is the gravity vector, and $\mathbf{u}$ is the vector of joint torques. When the end-effector is in contact with an external force $\mathbf{f}_{\text{ext}}$, the contact force is reflected into the joint space via the transpose Jacobian:

$$M(\mathbf{q}) \ddot{\mathbf{q}} + C(\mathbf{q}, \dot{\mathbf{q}}) \dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) = \mathbf{u} + J^T(\mathbf{q}) \mathbf{f}_{\text{ext}}$$

The transpose Jacobian J^T maps end-effector forces to joint torques: pushing on the end-effector of a robot arm generates torques at all joints downstream of the contact point.

If a force is applied at an intermediate link (not the end-effector), a *separate* Jacobian resolved at that link's contact point must be used. Multiple distributed contacts require summing the contributions of all individual Jacobians.

11.4. Manipulator Dynamics in Cartesian Space

For coupling kinematically dissimilar master and slave, dynamics must be expressed in **Cartesian (task) space**. The analytic Jacobian J_A relates end-effector velocity to joint velocity and is connected to the geometric Jacobian J by:

$$J_A = T_A^{-1}(\phi) J(\mathbf{q})$$

where T_A is a transformation matrix depending on the chosen orientation representation (e.g., Euler angles). After substituting the kinematic relationships (direct kinematics, first and second derivatives via J_A), the Cartesian-space model takes the form:

$$\tilde{M}(\mathbf{x}) \ddot{\mathbf{x}} + \tilde{C}(\mathbf{x}, \dot{\mathbf{x}}) \dot{\mathbf{x}} + \tilde{\mathbf{g}}(\mathbf{x}) = \mathbf{f}_c + \tilde{\mathbf{f}}_{\text{ext}}$$

where $\mathbf{f}_c = J_A^{-T} \mathbf{u}$ is the Cartesian control wrench and $\tilde{\mathbf{f}}_{\text{ext}} = T_A^{-1}(\phi) \mathbf{f}_{\text{ext}}$ is the external contact wrench (human force at the master; environment force at the slave).

11.5. Feedback Linearisation and Impedance Control

To simplify analysis, **feedback linearisation** in Cartesian space is applied. The nonlinear terms are cancelled by choosing the control law appropriately, reducing the closed-loop dynamics to a desired second-order integrator. It is then possible to select an **impedance model** with desired parameters M_d , B_d , K_d :

$$M_d \ddot{\tilde{\mathbf{x}}} + B_d \dot{\tilde{\mathbf{x}}} + K_d \tilde{\mathbf{x}} = \tilde{\mathbf{f}}_{\text{ext}},$$

where $\tilde{\mathbf{x}} = \mathbf{x} - \mathbf{x}_d$ is the Cartesian tracking error, further assuming the inertia matrix is **constant and diagonal** (see slides on teleoperation and on interaction control for the control laws and the choice of the impedance model parameters along with the information required for control implementation).

The underlying single-axis models used in all subsequent analysis are:

Master (single axis):

$$m_m \ddot{x}_m + b_m \dot{x}_m + k_m x_m = f_{Mc} + f_H$$

Slave (single axis):

$$m_s \ddot{x}_s + b_s \dot{x}_s + k_s x_s = f_{Sc} + f_E$$

where m_m/b_m and m_s/b_s are the scalar inertia and damping after feedback linearisation; f_{Mc}/f_{Sc} are controller-generated forces; f_H/f_E are external forces (human at master, environment at slave). During contact: $x_H = x_m$ and $x_e = x_s$.

11.6. Models in the Frequency Domain

Taking the Laplace transform, the **impedances** of all four subsystems are:

Subsystem	Impedance $Z(s)$	Notes
Human arm	$Z_H(s) = M_H s + B_H + K_H/s$	Full visco-inertial arm model
Remote environment	$Z_E(s) = M_E s + B_E + K_E/s$	Same structure
Master manipulator	$Z_M(s) = m_m s + b_m + k_m/s$	After feedback linearisation
Slave manipulator	$Z_S(s) = m_s s + b_s + k_s/s$	After feedback linearisation

12. Control

In principle any control methodology could be applied to the bilateral teleoperation problem. However, a universally accepted solution does not yet exist, and even agreeing on a performance criterion by which to compare architectures is non-trivial. The main design dimensions along which control schemes differ are:

- **Admittance vs. impedance formulation.** At each side (master or slave), the robot can be controlled as an impedance or as an admittance. Impedance-based schemes are more popular in practice because they do not necessarily require force sensors at the end-effector.
- **Information exchanged.** What signals are sent over the communication channel? Position only, force only, both, or hybrid signals? The choice affects both performance and robustness to delay.
- **Sensors required.** Force/torque sensors, joint encoders, vision-based force estimation, ... The sensor set determines cost, invasiveness (no cable through sterile drapes), and bandwidth.
- **Computational load and required bandwidth.** High-frequency force control loops demand faster clocks and lower-latency communication than slow position-only loops.

The specific control architectures (position–position, force–force, position–force, four-channel, etc.) will be formalised, compared, and analysed in Lecture 13.

13. Control Objectives

13.1. Stability

The teleoperation system is stable if every state variable (positions, velocities, internal controller states) remains **bounded** in response to bounded external inputs (forces/torques applied by the human operator and the environment).

From an energy perspective: a *passive* (energy-dissipating) system cannot amplify bounded inputs. Communication delays compound the problem, since they can cause a controller to keep *injecting* energy into the loop long after it should have stopped, leading to divergence even in systems that would be stable with zero delay.

13.2. Transparency

Transparency is a multi-faceted property. The operator should perceive a direct physical interaction with the remote environment during task execution. Concretely:

1. **Inertia/damping at master.** When *no force* is exerted on the slave, the inertia and damping perceived by the human at the master side should be low (ideally zero) so that free-space motion feels effortless.
2. **Tracking.** The slave should accurately reproduce the master displacements during free motion. Delays in the communication channel directly degrade this.
3. **Stiffness perception.** During *contact*, the stiffness (and more generally the impedance) perceived by the operator should equal the actual environmental impedance.
4. **Zero drift.** The position difference (drift) between master and slave should be zero during interaction.

As shown in Section 9.3, complete transparency requires $\lambda_v = \lambda_f = 1$ (telepresence), which conflicts with stability in the presence of delays. The design of a bilateral controller is therefore always an explicit **trade-off between transparency and stability**.

14. Preview: Two-Port Network Model (Lecture 13)

In the next lecture the system will be analysed using the **two-port network** formalism, borrowing the analogy with electrical networks:

Mechanical	Electrical analogy
Force f (effort variable)	Voltage V
Velocity $\dot{x}$ (flow variable)	Current I
Power $P = f\dot{x}$	Power $P = VI$
Mechanical impedance $Z = f/\dot{x}$	Electrical impedance $Z = V/I$

The two ports represent the human–master interface (port 1) and the slave–environment interface (port 2) respectively. The hybrid matrix H relating the port variables encodes both the stability and transparency properties of the closed-loop system, and provides a clean framework for comparing different control architectures.

15. Summary and Key Conceptual Threads

- **Teleoperation** extends human sensing and manipulation to a remote site using a master–slave architecture. **Telepresence** is the ideal of transparent bilateral interaction; **telerobotics** requires robotic devices on both sides.
- **Three control modalities:** direct bilateral, shared, supervisory. This course focuses on direct bilateral control without local slave autonomy.

- **Kinematic coupling:** equivalent devices couple at the joint level; dissimilar devices couple at the task (Cartesian) level with initial-offset compensation. Clutching handles workspace mismatch.
- **Scaling factors** λ_v and λ_f shape kinesthetic perception through the relations $P_m = \lambda_f/\lambda_v P_s$ and $Z_m = \lambda_v \lambda_f Z_s$. Only three of the four port-variable relationships are independent.
- **Transparency–stability trade-off:** $\lambda_v = \lambda_f = 1$ gives telepresence but exposes the system to instability with delays. The surgical choice $\lambda_v = \lambda_f < 1$ provides power attenuation and motion scaling at the cost of impedance under-estimation.
- **System models:** after feedback linearisation and the constant-diagonal-inertia approximation, all four subsystems reduce to scalar second-order systems dynamic impedances.
- **Control objectives:** BIBO stability and four-component transparency (inertia/damping, tracking, stiffness, zero drift).

Key topics for revision.

1. Definitions: teleoperation, telepresence, telerobotics and their mathematical content.
2. Components of a telemanipulator; the three control modalities.
3. Joint-level vs. task-level kinematic coupling; offset computation; clutching.
4. Scaling factors λ_v , λ_f : power formula $P_m = \lambda_f/\lambda_v P_s$, impedance formula $Z_m = \lambda_v \lambda_f Z_s$, and the independence constraint.
5. The telepresence condition ($\lambda_v = \lambda_f = 1$) and why it conflicts with stability when delays are present.
6. The surgical scaling choice $\lambda_v = \lambda_f < 1$: safety, precision, and the cost in transparency.
7. Second-order models of human, environment, master, and slave after feedback linearisation.
8. Stability and the four sub-requirements of transparency.

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